

# **Exchange Rate Depreciation and Inflation in Pakistan (2020–2025)**

## **A Descriptive and Correlation Analysis**

**Independent Research Project**

**Prepared By**

**Tooba Khalil**

**BS Economics**

**Government College University Lahore**

**July 2026**

|                                                                        |   |
|------------------------------------------------------------------------|---|
| Exchange Rate Depreciation and Inflation in Pakistan (2020–2025) ..... | 1 |
| A Descriptive and Correlation Analysis .....                           | 1 |
| Independent Research Project .....                                     | 1 |
| Abstract: .....                                                        | 3 |
| 1. Introduction: .....                                                 | 3 |
| 3. Research Questions .....                                            | 4 |
| 4. Data Source .....                                                   | 4 |
| 5. Methodology .....                                                   | 4 |
| 6. Results .....                                                       | 5 |
| 7. Discussion .....                                                    | 5 |
| 8. Conclusion .....                                                    | 5 |
| 9. Limitations .....                                                   | 6 |
| 10. Recommendations .....                                              | 6 |
| References : .....                                                     | 6 |

## **Abstract:**

The relationship between movement of exchange rate and inflation in Pakistan from January 2020 to June 2025 is explored in this study. The monthly figures of exchange rate USD/PKR and the inflation rate calculated based on Consumer Price Index (CPI) have been analyzed with the help of Microsoft Excel software. Descriptive statistics, correlation analysis and data visualization techniques are used to analyze trends and patterns of the two macro economic variables. The results show that there is a moderate positive relationship ( $r = 0.423$ ), meaning that an appreciation of the exchange rate is related to higher inflation in the period of study. The analysis also shows that there was a significant surge in inflation in 2022-2023, which then leveled off in 2024-2025. The project shows how data analytics can be used in the context of macroeconomic developments and gives insights into the recent inflationary trend in Pakistan.

**Keywords:** Pakistan, Inflation, Exchange Rate, CPI, Macroeconomics, Data Analysis, Correlation Analysis

## **1. Introduction:**

The two most relevant macro-economic indicators that impact economic stability are inflation and exchange rate fluctuation. Devaluation impacts the cost of foreign goods and raw materials in the domestic economy, and hence inflationary pressures. The macroeconomic issues faced by Pakistan were significant during the period 2020-2025 and comprised of currency depreciation, supply side shocks, and inflation of consumer prices. The link between the movement of exchange rates and inflation is critical to policy makers, researchers, and investors. This study aims to give a descriptive picture of these variables by using monthly data, and to check if there is a statistical relationship between exchange rate changes and inflation for the period in question.

## **2. Research Objective**

The objectives of this study are:

To analyze the trend of Pakistan's monthly exchange rate from 2020 to 2025.

To examine the trend of monthly inflation over the same period.

To calculate descriptive statistics for both variables.

To measure the correlation between exchange rate movements and inflation.

To visualize macroeconomic trends using an interactive dashboard.

### **3. Research Questions**

The study addresses the following research questions:

How did Pakistan's exchange rate change between 2020 and 2025?

How did inflation behave during the study period?

Is there a measurable relationship between exchange rate depreciation and inflation?

What insights can be obtained through descriptive statistics and data visualization?

### **4. Data Source**

The analysis uses publicly available secondary data collected from:

State Bank of Pakistan (SBP) – Monthly USD/PKR Exchange Rate

Pakistan Bureau of Statistics (PBS) – Consumer Price Index (CPI) Inflation

The dataset covers monthly observations from January 2020 to June 2025.

### **5. Methodology**

This research is a quantitative descriptive.

Data were collected, cleaned, and organized in the Microsoft Excel. Observations that were missing and variables that were not needed were eliminated in order to be consistent. The following analytical techniques were used:

Data cleansing and data conversion.

Monthly trend analysis

Annual average calculations

Descriptive statistics

Pearson correlation analysis

The use of charts and key performance indicators (KPIs) for dashboard visualization.

The dashboard provides a visual summary of exchange rate activity, inflation activity, averages of exchange rates over years and statistical measures.

## 6. Results

The descriptive analysis reveals that there was significant depreciation in the exchange rate of Pakistan in the study period. In 2020-2021 the exchange rate remained relatively stable, but in 2022 and 2023, it depreciated significantly. There was a relatively consistent trend in 2024–2025. Inflation also rose at a similar pace. Inflation has been rising slowly in 2021 and peaked in 2022-2023. Later, the rate of inflation started to fall consistently in 2024 and 2025.

The coefficient of correlation for the correlation analysis was 0.423, which shows there was a moderate positive correlation between the depreciation of the national currency and inflation. While correlation does not prove causation, the results indicate that there was a tendency for a higher inflation rate to be experienced during currency depreciation in the study period.

These macroeconomic patterns are also reflected in the dashboard visualisations, and give a intuitive understanding of changes over time.

## 7. Discussion

The results are consistent with the prevailing macroeconomic theory which indicates that exchange rate depreciation can lead to inflation by means of higher import prices and higher production costs. This relationship might have been intensified during the study period due to Pakistan's dependence on imported fuel, machinery and intermediate products.

But exchange rate fluctuations are not the sole factor that affects inflation. Other factors that affect inflation dynamics are global commodity prices, fiscal policies, monetary policies, energy costs and supply chain disruptions. It is important to note that the correlation observed is therefore considered as an association and not as a direct causal link.

## 8. Conclusion

This research paper explored the link between depreciation in the exchange rate and inflation in Pakistan for the period of January 2020-June 2025 on a month-on-month basis. The findings suggest that the inflation and exchange rate dynamics were similar in periods of economic crises. There was a moderate positive correlation ( $r = 0.423$ ) which indicated that exchange rate depreciation was positively correlated with inflation during the period of the study.

The dashboard and statistical analysis show that analytical tools that can be applied to the spreadsheet can effectively be used to explore macroeconomic relationships. Another important

aspect of this project is that it presents a good example of the importance of data visualization in conveying information about economic trends and changes to researchers and policy makers.

## 9. Limitations

This study has several limitations:

The analysis relies solely on secondary data.

Only two macroeconomic variables were examined.

Correlation analysis does not imply causality.

The study uses Microsoft Excel rather than advanced econometric software.

External economic shocks were not modeled explicitly.

## 10. Recommendations

Future research may extend this analysis by:

Incorporating additional macroeconomic variables such as interest rates, money supply, GDP growth, and unemployment.

Applying regression analysis and time-series econometric models.

Comparing Pakistan's experience with other developing economies.

Using software such as R, Python, Stata, or EViews for more advanced statistical analysis.

## References :

*Pakistan Bureau of Statistics. (2025). Consumer Price Index (CPI).* <https://www.pbs.gov.pk>

*State Bank of Pakistan. (2025). Exchange Rate Statistics.* <https://www.sbp.org.pk>